H2 Mathematics · Set 1 · Paper 1 · A-Level format · Worked solutions

100 marks

1. A sequence $u_1, u_2, u_3, \ldots$ is defined by

$u_1 = a$ and $u_{n+1} = u_n^2 - 6$ for $n \geq 1$,

where a is a constant.

(a) It is given that $u_3 = u_1$ and that the sequence is not constant. Find the possible values of a, giving your answers in exact form. [3]

(b) It is now given that $a > 0$. Find the exact value of $\sum_{r=1}^{25} u_r$. [2]

Solution:

(a) $u_2 = a^2 - 6$ and $u_3 = (a^2 - 6)^2 - 6 = a^4 - 12a^2 + 30$.

$u_3 = u_1 \Rightarrow a^4 - 12a^2 - a + 30 = 0$

The sequence is constant exactly when $u_2 = u_1$ (then every later term is the same), i.e. $a^2 - 6 = a$, $a^2 - a - 6 = 0$, $(a-3)(a+2) = 0$, $a = 3$ or $a = -2$. A constant sequence also has $u_3 = u_1$, so $a = 3$ and $a = -2$ are roots of the quartic and $a^2 - a - 6$ is a factor:

$a^4 - 12a^2 - a + 30 = (a^2 - a - 6)(a^2 + a - 5)$

✓ **Check:** $(a^2 - a - 6)(a^2 + a - 5) = a^4 + a^3 - 5a^2 - a^3 - a^2 + 5a - 6a^2 - 6a + 30 = a^4 - 12a^2 - a + 30$.

The sequence is not constant, so $a \neq 3$ and $a \neq -2$ (rejected: they give $u_2 = u_1$). Hence $a^2 + a - 5 = 0$,

$$a = \frac{-1 \pm \sqrt{1 + 20}}{2} = \frac{-1 \pm \sqrt{21}}{2}$$

(For these values $u_2 = a^2 - 6 = (5 - a) - 6 = -1 - a$, which differs from a since $a \neq -\frac{1}{2}$, so the sequence is indeed not constant.)

(b) $a > 0 \Rightarrow a = \dfrac{-1 + \sqrt{21}}{2}$

Since $a^2 = 5 - a$, $u_2 = a^2 - 6 = -1 - a = \frac{-1-\sqrt{21}}{2}$, and $u_3 = u_1$, so the terms alternate: $u_r = a$ for odd r and $u_r = -1 - a$ for even r. Each consecutive pair sums to $u_1 + u_2 = a + (-1 - a) = -1$.

$\sum_{r=1}^{25} u_r = 12(u_1 + u_2) + u_{25} = 12(-1) + a = -12 + \frac{-1+\sqrt{21}}{2} = \frac{\sqrt{21}-25}{2}$ (≈ -10.2).

(A calculator iteration from a rounded a is useless here: the 2-cycle is unstable and rounding error grows by a factor of about 20 every two terms, so the exact periodic argument is required.)

2. (a) An arithmetic series has first term a and common difference d. The sum of the first m terms of the series is equal to the sum of the first n terms, where m and n are positive integers and $m \neq n$. Show that the sum of the first $(m + n)$ terms of the series is zero. [2]

(b) A geometric series has first term 6 and its sum to infinity is 4. The sum of the first n terms of this series is S_n. Find the values of n for which S_n exceeds the sum to infinity by more than 0.01. [3]

Solution:

(a) Let S_k denote the sum of the first k terms, $S_k = \frac{k}{2}[2a + (k-1)d]$.

$$S_m = S_n \Rightarrow \frac{m}{2}[2a + (m-1)d]$$

$$= \frac{n}{2}[2a + (n-1)d]$$

$$\Rightarrow 2a(m-n) + d[m(m-1) - n(n-1)] = 0$$

Now $m(m-1) - n(n-1) = (m^2 - n^2) - (m-n) = (m-n)(m+n-1)$, so

$(m-n)[2a + (m+n-1)d] = 0$. [M1: equate the sums and take out the factor $(m-n)$]

Since $m \neq n$, $m - n \neq 0$, hence $2a + (m+n-1)d = 0$.

Therefore $S_{m+n} = \frac{m+n}{2}[2a + (m+n-1)d] = \frac{m+n}{2}(0) = 0$. [A1: $m \neq n$ used and the bracket recognised as that of S_{m+n}]

✓ **Check:** $a = 41, d = -2, m = 12, n = 30$ gives $S_{12} = S_{30} = 360$ and $S_{42} = 0$.

(b) Let the common ratio be r. $\frac{6}{1-r} = 4 \Rightarrow 1 - r = \frac{3}{2} \Rightarrow r = -\frac{1}{2}$ (and $|r| < 1$, so the series converges; the sum to infinity is less than the first term because $r < 0$). [B1]

$S_n = \frac{6\left[1-\left(-\frac{1}{2}\right)^n\right]}{1-\left(-\frac{1}{2}\right)} = 4\left[1 - \left(-\frac{1}{2}\right)^n\right]$, so $S_n - 4 = -4\left(-\frac{1}{2}\right)^n$.

Required: $-4\left(-\frac{1}{2}\right)^n > 0.01$. [M1]

If n is even, $\left(-\frac{1}{2}\right)^n > 0$, so $S_n - 4 < 0$: S_n is below the sum to infinity and no even n works.

If n is odd, $\left(-\frac{1}{2}\right)^n = -\left(\frac{1}{2}\right)^n$, so $S_n - 4 = 4\left(\frac{1}{2}\right)^n > 0.01 \Leftrightarrow \left(\frac{1}{2}\right)^n > \frac{1}{400} \Leftrightarrow 2^n < 400 \Leftrightarrow n \leq 8$ (since $2^8 = 256$ and $2^9 = 512$).

Hence $n = 1,3,5,7$. [A1]

✓ **Check:** $S_7 = 4 + \frac{4}{128} = 4.03125 > 4.01$; $S_9 = 4 + \frac{4}{512} = 4.0078125 < 4.01$; $S_8 = 3.984375 < 4$.

3. The function f is defined by

$$\mathrm{f}: x \mapsto \tan^{-1}\left(\frac{x+\sqrt{3}}{1-\sqrt{3}\,x}\right) - \tan^{-1} x, \quad x \in \mathbb{R},\ x \neq \frac{1}{\sqrt{3}}.$$

(a) Show that $\mathrm{f}'(x) = 0$. [3]

(b) Hence find the range of f, giving your answer in exact form. [3]

Solution:

(a) Let $u = \frac{x+\sqrt{3}}{1-\sqrt{3}\,x}$.

By the quotient rule, $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{(1)(1-\sqrt{3}\,x)-(x+\sqrt{3})(-\sqrt{3})}{(1-\sqrt{3}\,x)^2} = \frac{1-\sqrt{3}\,x+\sqrt{3}\,x+3}{(1-\sqrt{3}\,x)^2} = \frac{4}{(1-\sqrt{3}\,x)^2}$. M1

$1 + u^2 = \frac{(1-\sqrt{3}\,x)^2+(x+\sqrt{3})^2}{(1-\sqrt{3}\,x)^2} = \frac{1-2\sqrt{3}\,x+3x^2+x^2+2\sqrt{3}\,x+3}{(1-\sqrt{3}\,x)^2} = \frac{4(1+x^2)}{(1-\sqrt{3}\,x)^2}$. M1

So $\frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1} u = \frac{1}{1+u^2}\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{(1-\sqrt{3}\,x)^2}{4(1+x^2)} \cdot \frac{4}{(1-\sqrt{3}\,x)^2} = \frac{1}{1+x^2}$.

Hence $\mathrm{f}'(x) = \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0$ for every x in the domain of f (shown). A1

(Alternative: $\tan\left(\frac{\pi}{3} + \tan^{-1} x\right) = \frac{\sqrt{3}+x}{1-\sqrt{3}\,x}$, so $\tan^{-1} u$ differs from $\frac{\pi}{3} + \tan^{-1} x$ by a constant multiple of π on each interval, and its derivative is $\frac{1}{1+x^2}$.)

(b) Since $f'(x) = 0$, f is constant on each interval of its domain. The domain is NOT one interval: it is $x < \frac{1}{\sqrt{3}}$ together with $x > \frac{1}{\sqrt{3}}$, and f need not take the same constant on both. (As $x \to \frac{1}{\sqrt{3}}$ from below, $u \to +\infty$ and $\tan^{-1} u \to \frac{\pi}{2}$; from above, $u \to -\infty$ and $\tan^{-1} u \to -\frac{\pi}{2}$, so f jumps by π there.) M1

For $x < \frac{1}{\sqrt{3}}$: take $x = 0$. $f(0) = \tan^{-1}\sqrt{3} - \tan^{-1} 0 = \frac{\pi}{3}$, so $f(x) = \frac{\pi}{3}$ for all $x < \frac{1}{\sqrt{3}}$. A1

For $x > \frac{1}{\sqrt{3}}$: take $x = \sqrt{3}$. Then $u = \frac{2\sqrt{3}}{1-3} = -\sqrt{3}$, so $f(\sqrt{3}) = \tan^{-1}(-\sqrt{3}) - \tan^{-1}\sqrt{3} = -\frac{\pi}{3} - \frac{\pi}{3} = -\frac{2\pi}{3}$, so $f(x) = -\frac{2\pi}{3}$ for all $x > \frac{1}{\sqrt{3}}$.

(Or: as $x \to \infty$, $u \to -\frac{1}{\sqrt{3}}$ and $\tan^{-1} x \to \frac{\pi}{2}$, so $f(x) \to -\frac{\pi}{6} - \frac{\pi}{2} = -\frac{2\pi}{3}$.)

Range of f: $R_f = \left\{-\frac{2\pi}{3}, \frac{\pi}{3}\right\}$. A1

A candidate who evaluates at one point only and gives $R_f = \left\{\frac{\pi}{3}\right\}$ scores at most M1 A1 A0.

(Check without (a): $\tan f(x) = \frac{u-x}{1+ux} = \frac{\frac{\sqrt{3}(1+x^2)}{(1-\sqrt{3}x)}}{\frac{(1+x^2)}{(1-\sqrt{3}x)}} = \sqrt{3}$ and $-\pi < f(x) < \pi$, so $f(x) \in \left\{\frac{\pi}{3}, -\frac{2\pi}{3}\right\}$; both values occur. Numerically $f(-5) = 1.0472$, $f(5) = -2.0944$.)

4. Do not use a calculator in answering this question.

On an Argand diagram, the points A and C represent the complex numbers $1 + 2i$ and $5 + 4i$ respectively. A and C are opposite vertices of a square $ABCD$, and the vertices A, B, C, D are in anticlockwise order.

(a) Find the complex numbers represented by B and D. [3]

(b) Show that the complex numbers represented by A, B, C and D are the roots of an equation of the form $(z - p)^4 = q$, where p and q are complex numbers to be found. [3]

(c) Hence find the four roots of the equation $w^4 = q$, and the square roots of q. [2]

Solution:

(a) The diagonals of a square are equal and bisect each other at right angles, so the centre M of the square is the midpoint of AC: $m = \frac{1}{2}[(1 + 2i) + (5 + 4i)] = 3 + 3i$. M1

$\overrightarrow{MA}$ represents $a - m = (1 + 2i) - (3 + 3i) = -2 - i$. MA, MB, MC, MD are equal in length and, as the vertices are in anticlockwise order, each is the one before turned anticlockwise through $\frac{\pi}{2}$ about M.

Multiplication by i is an anticlockwise rotation through $\frac{\pi}{2}$, so $b - m = i(a - m)$ and $d - m = -i(a - m)$. M1

$b - m = i(-2 - i) = 1 - 2i$, so B represents $4 + i$; $d - m = -i(-2 - i) = -1 + 2i$, so D represents $2 + 5i$. A1

✓ **Check:** $c - m = i(b - m) = i(1 - 2i) = 2 + i$, so $c = 5 + 4i$ as given; $|AB| = |3 - i| = \sqrt{10} = |BC| = |1 + 3i|$; going $A(1,2) \to B(4,1) \to C(5,4) \to D(2,5)$ is anticlockwise.) The two answers interchanged (clockwise order) score M1 M1 A0.

(b) Take $p = 3 + 3\mathrm{i}$, the centre of the square, and let $u = a - p = -2 - \mathrm{i}$. By part (a),
$a - p = u$, $b - p = \mathrm{i}u$, $c - p = \mathrm{i}^2 u = -u$, $d - p = \mathrm{i}^3 u = -\mathrm{i}u$. M1
Since $\mathrm{i}^4 = 1$, $(\mathrm{i}^k u)^4 = (\mathrm{i}^4)^k u^4 = u^4$ for $k = 0,1,2,3$, so each of a, b, c, d satisfies $(z - p)^4 = u^4$.
$u^2 = (-2 - \mathrm{i})^2 = 4 + 4\mathrm{i} + \mathrm{i}^2 = 3 + 4\mathrm{i}$, so $u^4 = (3 + 4\mathrm{i})^2 = 9 + 24\mathrm{i} + 16\mathrm{i}^2 = -7 + 24\mathrm{i}$. Hence $q = -7 + 24\mathrm{i}$. M1
a, b, c, d are four distinct roots of $(z - 3 - 3\mathrm{i})^4 = -7 + 24\mathrm{i}$, a polynomial equation of degree 4, which has at most four roots; so they are exactly its roots, with $p = 3 + 3\mathrm{i}$ and $q = -7 + 24\mathrm{i}$ (shown). A1
(A candidate who evaluates each fourth power separately, e.g. $(1 - 2\mathrm{i})^2 = -3 - 4\mathrm{i}$ and $(-3 - 4\mathrm{i})^2 = -7 + 24\mathrm{i}$, earns full marks. The values are forced: $(z - p)^4 - q = z^4 - 4pz^3 + \cdots$, and comparing with $(z - a)(z - b)(z - c)(z - d)$ gives $4p = a + b + c + d = 12 + 12\mathrm{i}$, so $p = 3 + 3\mathrm{i}$ and then $q = (a - p)^4$. Geometrically: if z_0 is a root of $(z - p)^4 = q$ then so is $p + \mathrm{i}(z_0 - p)$, so the roots form a square centred at p, and p must be the centre M.)

(c) Put $w = z - p$: $w^4 = q$ is $(z - p)^4 = q$, so its roots are $a - p$, $b - p$, $c - p$, $d - p$, i.e. $w = -2 - \mathrm{i},\ 1 - 2\mathrm{i},\ 2 + \mathrm{i},\ -1 + 2\mathrm{i}$. B1
If $w^4 = q$ then $(w^2)^2 = q$, so w^2 is a root of $s^2 = q$. From the roots above, $(-2 - \mathrm{i})^2 = (2 + \mathrm{i})^2 = 3 + 4\mathrm{i}$ and $(1 - 2\mathrm{i})^2 = (-1 + 2\mathrm{i})^2 = -3 - 4\mathrm{i}$. A quadratic equation has at most two roots, so $s = 3 + 4\mathrm{i}$ or $s = -3 - 4\mathrm{i}$. B1
✓ **Check:** $(3 + 4\mathrm{i})^2 = 9 + 24\mathrm{i} - 16 = -7 + 24\mathrm{i}$.

5. The function f is defined by $\mathrm{f}: x \mapsto x + \frac{1}{x}$, $x \in \mathbb{R}$, $x \neq 0$.

(a) Use calculus to find the range of f. [2]

(b) Hence find the exact values of the real roots of the equation $x^4 - 4x^3 + 5x^2 - 4x + 1 = 0$, explaining why the equation has no other real roots. [4]

(c) Find the set of values of the constant k for which the equation $x^4 - 4x^3 + kx^2 - 4x + 1 = 0$ has four distinct real roots. [2]

Solution:

(a) $\mathrm{f}'(x) = 1 - \frac{1}{x^2}$. $\mathrm{f}'(x) = 0 \Rightarrow x^2 = 1 \Rightarrow x = 1$ or $x = -1$.
$\mathrm{f}''(x) = \frac{2}{x^3}$: $\mathrm{f}''(1) = 2 > 0$, so $(1,2)$ is a minimum point; $\mathrm{f}''(-1) = -2 < 0$, so $(-1,-2)$ is a maximum point. [M1: stationary points $x = \pm 1$ with values ± 2]
For $x > 0$: f is continuous, $\mathrm{f}(x) \to \infty$ as $x \to 0^+$ and as $x \to \infty$, and its only stationary point is the minimum at $(1,2)$, so f takes every value in $[2, \infty)$ and no value below 2.
For $x < 0$: $\mathrm{f}(-x) = -\mathrm{f}(x)$ (equivalently $\mathrm{f}(x) \to -\infty$ as $x \to 0^-$ and as $x \to -\infty$, with maximum -2 at $x = -1$), so f takes every value in $(-\infty, -2]$ and no value above -2.
Range of $\mathrm{f} = (-\infty, -2] \cup [2, \infty)$. [A1]

(b) $x = 0$ is not a root (the constant term is 1), so divide by x^2:

$$x^2 - 4x + 5 - \frac{4}{x} + \frac{1}{x^2} = 0 \Rightarrow \left(x^2 + \frac{1}{x^2}\right) - 4\left(x + \frac{1}{x}\right) + 5 = 0$$

Let $u = \mathrm{f}(x) = x + \frac{1}{x}$. Then $u^2 = x^2 + 2 + \frac{1}{x^2}$, so $x^2 + \frac{1}{x^2} = u^2 - 2$ and $u^2 - 2 - 4u + 5 = 0$, i.e. $u^2 - 4u + 3 = 0$. [M1: divide by x^2 and use $x^2 + \frac{1}{x^2} = u^2 - 2$ to reach a quadratic in u]

$(u - 1)(u - 3) = 0 \Rightarrow u = 1$ or $u = 3$. [A1]

$x + \frac{1}{x} = 1$ has no real solution, because 1 is not in the range of f found in part (a) (it lies strictly between -2 and 2). [B1: the rejection with its reason]

$$x + \frac{1}{x} = 3 \Rightarrow x^2 - 3x + 1 = 0 \Rightarrow x = \frac{3 \pm \sqrt{5}}{2}$$

Every real root of the quartic gives $u = 1$ or $u = 3$, so the only real roots are $x = \frac{3-\sqrt{5}}{2}$ and $x = \frac{3+\sqrt{5}}{2}$ (approximately 0.382 and 2.62). [A1]

✓ **Check:** $x^4 - 4x^3 + 5x^2 - 4x + 1 = (x^2 - 3x + 1)(x^2 - x + 1)$. The factor theorem finds no rational root: $x = 1$ gives -1 and $x = -1$ gives 15.

(c) As in (b), $x = 0$ is not a root; dividing by x^2 and putting $u = x + \frac{1}{x}$ gives $u^2 - 2 - 4u + k = 0$, i.e. $u^2 - 4u + (k - 2) = 0$.

For a real u, $x + \frac{1}{x} = u \Leftrightarrow x^2 - ux + 1 = 0$, whose discriminant is $u^2 - 4$: two distinct real x when $|u| > 2$, one ($x = \pm 1$) when $|u| = 2$, none when $|u| < 2$ (as the range in (a) says). Different values of u give different values of x.

So the quartic has four distinct real roots ⇔ the quadratic in u has two distinct real roots, each with $|u| > 2$. The roots of $u^2 - 4u + (k - 2) = 0$ sum to 4, so they cannot both exceed 2 and cannot both be less than -2. Hence one root must be less than -2 and the other greater than 2: with $g(u) = u^2 - 4u + k - 2$, $g(-2) < 0$ and $g(2) < 0$. [M1]

$g(-2) = 4 + 8 + k - 2 = k + 10 < 0 \Rightarrow k < -10$; $g(2) = 4 - 8 + k - 2 = k - 6 < 0 \Rightarrow k < 6$.

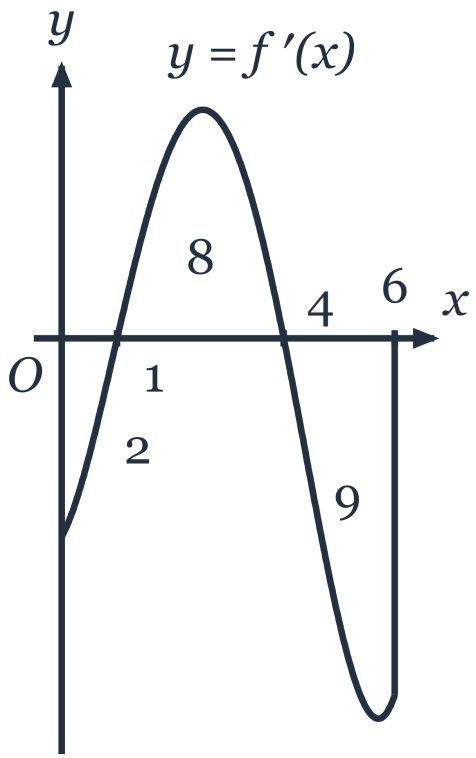
Both hold ⇔ $k < -10$. Set of values: $\{k \in \mathbb{R}: k < -10\}$. [A1]

(Equivalently: the roots are $u = 2 \pm \sqrt{6 - k}$; for $k < 6$ the larger always exceeds 2, and the smaller is below $-2 \Leftrightarrow \sqrt{6 - k} > 4 \Leftrightarrow k < -10$. At $k = -10$ the root $u = -2$ gives only $x = -1$, so there are three distinct real roots and the inequality is strict; for $-10 < k < 6$ there are exactly two, as in part (b) where $k = 5$; $k = 6$ gives only $x = 1$; $k > 6$ gives none.)

(Accepted alternative: $k = -\frac{x^4 - 4x^3 - 4x + 1}{x^2}$, whose graph has a maximum point $(-1, -10)$ for $x < 0$ and a maximum point $(1,6)$ for $x > 0$; the line $y = k$ meets it four times exactly when $k < -10$.)

Mark scheme notes (moderation, 27 Sep 2026). (b) Accepted alternative without part (a): matching coefficients in $(x^2 + ax + 1)(x^2 + bx + 1)$ with $a + b = -4$, $ab = 3$ gives $(x^2 - 3x + 1)(x^2 - x + 1)$; the second factor is rejected by its discriminant -3 — a complete answer with the explanation asked for; the B1 for the rejection with its reason is paid for the discriminant even when the range of f is not cited. (c) In the alternative route $k = 6 - (x - \frac{1)^4}{x^2}$, the point $(1,6)$ is a MAXIMUM for $x > 0$ (and $(-1, -10)$ a maximum for $x < 0$); each branch falls to $-\infty$ at both ends, so $y = k$ meets the graph four times exactly when $k < -10$.

6. The diagram shows the graph of $y = \mathrm{f}'(x)$ for $0 \leq x \leq 6$. The graph meets the x-axis only at $x = 1$ and $x = 4$, and the areas of the three regions between the graph and the x-axis are 2, 8 and 9, as shown. It is given that $\mathrm{f}(0) = 4$.



(a) State the x-coordinates of the stationary points of the curve $y = \mathrm{f}(x)$ for $0 \leq x \leq 6$ and determine, with a reason, the nature of each. [2]

(b) Find the greatest and least values of $\mathrm{f}(x)$ for $0 \leq x \leq 6$. [3]

(c) Find the coordinates of the stationary points of the curve $y = 3 - 2\mathrm{f}\left(\frac{1}{2}x\right)$ for $0 \leq x \leq 12$, and determine, with a reason, the nature of each. [3]

Solution:

(a) From the diagram, $\mathrm{f}'(x) = 0$ only at $x = 1$ and $x = 4$, so the stationary points of $y = \mathrm{f}(x)$ are at $x = 1$ and $x = 4$.

At $x = 1$ the graph of f' passes from below the x-axis to above it: $\mathrm{f}'(x)$ changes from negative to positive, so $x = 1$ gives a MINIMUM point. B1

At $x = 4$ it passes from above to below: $\mathrm{f}'(x)$ changes from positive to negative, so $x = 4$ gives a MAXIMUM point. B1

(b) f is an antiderivative of f′, so $\int_a^b \mathrm{f}'(x)\,\mathrm{d}x = \mathrm{f}(b) - \mathrm{f}(a)$; where the graph of f′ lies below the x-axis the integral is MINUS the area. M1

$\mathrm{f}(1) - \mathrm{f}(0) = \int_0^1 \mathrm{f}'(x)\,\mathrm{d}x = -2$, so $\mathrm{f}(1) = 4 - 2 = 2$.

$\mathrm{f}(4) - \mathrm{f}(1) = \int_1^4 \mathrm{f}'(x)\,\mathrm{d}x = 8$, so $\mathrm{f}(4) = 2 + 8 = 10$.

$\mathrm{f}(6) - \mathrm{f}(4) = \int_4^6 \mathrm{f}'(x)\,\mathrm{d}x = -9$, so $\mathrm{f}(6) = 10 - 9 = 1$. A1 (all three)

f is decreasing on $0 \le x \le 1$, increasing on $1 \le x \le 4$ and decreasing on $4 \le x \le 6$, so its greatest and least values on $0 \le x \le 6$ are among $\mathrm{f}(0) = 4$, $\mathrm{f}(1) = 2$, $\mathrm{f}(4) = 10$ and $\mathrm{f}(6) = 1$.

Greatest value $= 10$ (at $x = 4$). Least value $= 1$ (at the endpoint $x = 6$) — lower than the local minimum value $\mathrm{f}(1) = 2$. A1

(A candidate who adds the areas without their signs, e.g. $\mathrm{f}(1) = 6$ or $\mathrm{f}(6) = 19$, scores M0. A candidate who gives the least value as 2, the local minimum, scores A0 for the last mark.)

(c) Let $y = 3 - 2\mathrm{f}\left(\frac{1}{2}x\right)$, $0 \le x \le 12$. Then $\frac{\mathrm{d}y}{\mathrm{d}x} = -2 \cdot \frac{1}{2}\mathrm{f}'\left(\frac{1}{2}x\right) = -\mathrm{f}'\left(\frac{1}{2}x\right)$.

$\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \Leftrightarrow \mathrm{f}'\left(\frac{1}{2}x\right) = 0 \Leftrightarrow \frac{1}{2}x = 1$ or $\frac{1}{2}x = 4 \Leftrightarrow x = 2$ or $x = 8$. B1

At $x = 2$: $y = 3 - 2\mathrm{f}(1) = 3 - 4 = -1$. At $x = 8$: $y = 3 - 2\mathrm{f}(4) = 3 - 20 = -17$. The stationary points are $(2, -1)$ and $(8, -17)$. B1

Nature: as x increases through 2, $\frac{1}{2}x$ increases through 1, where f′ changes from negative to positive, so $\frac{\mathrm{d}y}{\mathrm{d}x} = -\mathrm{f}'\left(\frac{1}{2}x\right)$ changes from positive to negative: $(2, -1)$ is a MAXIMUM point. As x increases through 8, f′ changes from positive to negative, so $\frac{\mathrm{d}y}{\mathrm{d}x}$ changes from negative to positive: $(8, -17)$ is a MINIMUM point. B1

(Equivalently, by transformations: a stretch of factor 2 parallel to the x-axis takes the minimum $(1,2)$ and the maximum $(4,10)$ of $y = \mathrm{f}(x)$ to $(2,2)$ and $(8,10)$; then $y \mapsto 3 - 2y$ — a stretch of factor 2 parallel to the y-axis, a reflection in the x-axis and a translation of 3 units in the positive y-direction — gives $(2, -1)$ and $(8, -17)$, and the reflection turns the minimum into a maximum and the maximum into a minimum.)

✓ **Check:** the endpoints $x = 0$ and $x = 12$ give $y = 3 - 2(4) = -5$ and $y = 3 - 2(1) = 1$; they are not stationary points, since $\mathrm{f}'(0) \neq 0$ and $\mathrm{f}'(6) \neq 0$.

(Verification of the figure: with $\mathrm{f}'(x) = \frac{(x-1)(x-4)(1285x^2 - 7255x - 9672)}{10800}$, exact integration gives $\int_0^1 \mathrm{f}' = -2$, $\int_1^4 \mathrm{f}' = 8$, $\int_4^6 \mathrm{f}' = -9$; the quadratic factor has roots about -1.11 and 6.76, so f′ vanishes in $[0,6]$ only at $x = 1$ and $x = 4$; and $\mathrm{f}(x) = 4 + \int_0^x \mathrm{f}'$ has minimum 1 at $x = 6$ and maximum 10 at $x = 4$ on $[0,6]$.)

7. A straight rail AB of length $2a$ m is fixed at both ends. When the rail is heated, its length increases to $2a(1 + k)$ m, where k is a positive constant. The ends A and B remain $2a$ m apart, so the rail bows upwards into an arc of a circle of radius r m. This arc subtends an angle of 2θ radians at the centre of the circle.

(a) Show that $\theta = (1 + k)\sin\theta$. [2]

(b) It is given that k is small, so that θ is also small. Find an approximation for θ in terms of k. [3]

(c) Find, in terms of a and θ, the greatest height of the arc above the line AB, and deduce that this height is approximately $\frac{1}{2}a\theta$. [2]

(d) A rail of length 60 m, fixed at both ends, increases in length by 1 cm. Estimate the greatest height of the arc above the line joining the fixed ends, and comment on the size of this height compared with the increase in the length of the rail. [2]

Solution:

(a) Let O be the centre of the circle and M the midpoint of AB.

The arc AB subtends 2θ at O and has the new length of the rail: $r(2\theta) = 2a(1+k)$, so $r\theta = a(1+k)$. [B1]

OM is perpendicular to AB and bisects angle AOB, so angle $AOM = \theta$ and $AM = r\sin\theta = a$ (the chord AB is still $2a$).

Dividing, $\frac{r\theta}{r\sin\theta} = \frac{a(1+k)}{a}$, so $\frac{\theta}{\sin\theta} = 1+k$, i.e. $\theta = (1+k)\sin\theta$. [B1, AG]

(b) The first approximation for small θ, $\sin\theta \approx \theta$, gives $\theta = (1+k)\theta$, i.e. $k\theta = 0$, so $\theta = 0$ since $k > 0$. That is a straight rail, impossible because the rail is now longer than AB: the first-order terms cancel and this approximation gives no information about θ. The next term of the Maclaurin series must be kept: $\sin\theta \approx \theta - \frac{\theta^3}{6}$. [M1: the θ^3 term used, with or without the failed first attempt shown]

$\theta \approx (1+k)\left(\theta - \frac{\theta^3}{6}\right)$. As $\theta \neq 0$, divide by θ: $1 \approx (1+k)\left(1 - \frac{\theta^2}{6}\right)$. [M1: cancel θ and rearrange for θ^2]

$$1 - \frac{\theta^2}{6} \approx \frac{1}{1+k} \Rightarrow \frac{\theta^2}{6} \approx 1 - \frac{1}{1+k} = \frac{k}{1+k} \Rightarrow \theta^2 \approx \frac{6k}{1+k}$$

Since k is small, $1+k \approx 1$, so $\theta^2 \approx 6k$ and, as $\theta > 0$, $\theta \approx \sqrt{6k}$. [A1; accept $\sqrt{\frac{6k}{1+k}}$]

(Alternative: $\frac{\theta}{\sin\theta} = \left(1 - \frac{\theta^2}{6} + \cdots\right)^{-1} \approx 1 + \frac{\theta^2}{6}$, so $1 + \frac{\theta^2}{6} \approx 1 + k$ and $\theta^2 \approx 6k$.)

(c) $OM = r\cos\theta$. The rail bows upwards and θ is small, so O lies below AB and the highest point P of the arc lies on OM produced, with $OP = r$. The greatest height is

$h = MP = r - r\cos\theta = r(1-\cos\theta) = \frac{a(1-\cos\theta)}{\sin\theta}$, using $r = \frac{a}{\sin\theta}$ from (a). [B1; accept $a\tan\frac{1}{2}\theta$ or $a\operatorname{cosec}\theta - a\cot\theta$]

For small θ, $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ and $\sin\theta \approx \theta$, so $h \approx \frac{a\cdot\frac{1}{2}\theta^2}{\theta} = \frac{1}{2}a\theta$. [B1, AG]

(Here the leading terms do not cancel, so the first terms of the series are enough; taking $\cos\theta \approx 1$ would give $h \approx 0$, the same trap as in (b).)

In terms of k: $h \approx \frac{1}{2}a\sqrt{6k} = a\sqrt{\frac{3k}{2}}$.

(d) $2a = 60$, so $a = 30$. The increase in length is $2ak = 0.01$ m, so $k = \frac{0.01}{60} = \frac{1}{6000}$.

$\theta \approx \sqrt{6k} = \sqrt{0.001} \approx 0.03162$ (small, so the approximations are justified).

Greatest height $\approx \frac{1}{2}a\theta = 15 \times 0.03162 \approx 0.474$ m, about 47 cm. [B1]

✓ **Check:** solving $\theta = \left(1 + \frac{1}{6000}\right)\sin\theta$ on the graphing calculator gives $\theta = 0.031621$ and $h = 30\tan\frac{1}{2}\theta = 0.4744$ m; the radius is about 949 m.

Comment: a 1 cm increase in the length of a 60 m rail lifts its middle by about 47 cm, nearly 50 times the expansion itself. Because the rise is about $\frac{1}{2}a\sqrt{6k}$, it grows like $\sqrt{k}$ rather than k, so even a very small expansion produces a large bow; a rail fixed at both ends with no room to expand would buckle dangerously, which is why gaps are left between rails for expansion. [B1: the height is large compared with the 1 cm expansion, with a consequence in context]

8. The curve C has parametric equations $x = t + \sin t$, $y = \sin t$, for $0 \leq t \leq \pi$. The curve C meets the x-axis at the origin O and at the point F.

(a) Show that, for $0 \leq t < \pi$, $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\cos t}{1+\cos t}$. [2]

(b) The line $y = h$, where h is a constant such that $0 < h < 1$, meets C at the points A and B, where the x-coordinate of A is less than the x-coordinate of B. The tangents to C at A and at B are perpendicular. Find the exact value of h and the exact coordinates of B. [4]

(c) Find the exact area of the region bounded by the tangent to C at B, the x-axis and the arc of C between B and F. [5]

Solution:

(a) $\frac{\mathrm{d}x}{\mathrm{d}t} = 1 + \cos t$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = \cos t$. For $0 \leq t < \pi$, $1 + \cos t \neq 0$, so $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\frac{\mathrm{d}y}{\mathrm{d}t}}{\frac{\mathrm{d}x}{\mathrm{d}t}} = \frac{\cos t}{1+\cos t}$. [B1 both derivatives; A1 shown]

(b) A point of C with $y = h$ has $\sin t = h$. For $0 < h < 1$ this has exactly two roots in $0 \leq t \leq \pi$: $t = \alpha$ and $t = \pi - \alpha$, where $0 < \alpha < \frac{\pi}{2}$, because $\sin(\pi - \alpha) = \sin \alpha$. Since $\frac{\mathrm{d}x}{\mathrm{d}t} = 1 + \cos t > 0$ for $0 \leq t < \pi$, x increases with t, so A has parameter α and B has parameter $\pi - \alpha$. [M1]

Gradient at A: $\frac{\cos \alpha}{1+\cos \alpha}$. Gradient at B: $\frac{\cos(\pi-\alpha)}{1+\cos(\pi-\alpha)} = \frac{-\cos \alpha}{1-\cos \alpha}$. [M1]

Perpendicular tangents: $\frac{\cos \alpha}{1+\cos \alpha} \times \frac{-\cos \alpha}{1-\cos \alpha} = -1 \Rightarrow \frac{\cos^2 \alpha}{1-\cos^2 \alpha} = 1 \Rightarrow \cos^2 \alpha = \sin^2 \alpha \Rightarrow \tan^2 \alpha = 1$. As $0 < \alpha < \frac{\pi}{2}$, $\tan \alpha = 1$, so $\alpha = \frac{\pi}{4}$. [M1] (Equivalently, with $\sin \alpha = h$: $\frac{1-h^2}{h^2} = 1$, $h^2 = \frac{1}{2}$.)

$h = \sin \frac{\pi}{4} = \frac{\sqrt{2}}{2}$. B has parameter $\frac{3\pi}{4}$: $x = \frac{3\pi}{4} + \sin \frac{3\pi}{4} = \frac{3\pi}{4} + \frac{\sqrt{2}}{2}$, $y = \frac{\sqrt{2}}{2}$, so $B = \left(\frac{3\pi}{4} + \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$. [A1] (Check: the gradients are $\sqrt{2} - 1$ at A and $-(\sqrt{2} + 1)$ at B, product $-(2 - 1) = -1$.)

(c) F: $\sin t = 0$ with $t \neq 0$ gives $t = \pi$, so $F = (\pi, 0)$.

Gradient at B: $\frac{-\frac{\sqrt{2}}{2}}{1-\frac{\sqrt{2}}{2}} = \frac{-\sqrt{2}}{2-\sqrt{2}} = -\frac{\sqrt{2}(2+\sqrt{2})}{2} = -(\sqrt{2} + 1)$.

Tangent at B: $y - \frac{\sqrt{2}}{2} = -(\sqrt{2} + 1)\left(x - \frac{3\pi}{4} - \frac{\sqrt{2}}{2}\right)$. [M1]

At $y = 0$: $x - \frac{3\pi}{4} - \frac{\sqrt{2}}{2} = \frac{\frac{\sqrt{2}}{2}}{\sqrt{2}+1} = \frac{\sqrt{2}}{2}(\sqrt{2} - 1) = 1 - \frac{\sqrt{2}}{2}$, so the tangent meets the x-axis at $S\left(\frac{3\pi}{4} + 1,\ 0\right)$. [A1] Since $1 > \frac{\pi}{4}$, $\frac{3\pi}{4} + 1 > \pi$, so S lies beyond F: the region is bounded above by the segment BS of the tangent and below by the arc BF of C and the segment FS of the x-axis. (The tangent lies above C, since $\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = -\frac{\sin t}{(1+\cos t)^3} \leq 0$.)

Let $N = \left(\frac{3\pi}{4} + \frac{\sqrt{2}}{2},\ 0\right)$ be the point of the x-axis directly below B. Area = area of triangle BNS − area between the arc BF and the x-axis. [M1]

Triangle BNS: $NS = \left(\frac{3\pi}{4} + 1\right) - \left(\frac{3\pi}{4} + \frac{\sqrt{2}}{2}\right) = 1 - \frac{\sqrt{2}}{2}$ and $BN = \frac{\sqrt{2}}{2}$, so its area is $\frac{1}{2}\left(1 - \frac{\sqrt{2}}{2}\right)\frac{\sqrt{2}}{2} = \frac{\sqrt{2}-1}{4}$.

Area under arc $BF = \int y \,\mathrm{d}x = \int_{\frac{3\pi}{4}}^{\pi} \sin t\,(1 + \cos t)\,\mathrm{d}t = \left[-\cos t + \frac{1}{2}\sin^2 t\right]_{\frac{3\pi}{4}}^{\pi}$ [M1] $= (1 + 0) - \left(\frac{\sqrt{2}}{2} + \frac{1}{4}\right) = \frac{3}{4} - \frac{\sqrt{2}}{2}$.

Required area $= \frac{\sqrt{2}-1}{4} - \left(\frac{3}{4} - \frac{\sqrt{2}}{2}\right) = \frac{3\sqrt{2}-4}{4}\left(= \frac{3\sqrt{2}}{4} - 1 \approx 0.0607\right)$. [A1]

9. For $0 \leq k \leq \ln 3$, the function J is defined by $\mathrm{J}(k) = \int_1^3 |\ln x - k| \, \mathrm{d}x$.

(a) Find $\int \ln x \, \mathrm{d}x$. [2]

(b) Find $\mathrm{J}(k)$ in terms of k. [4]

(c) Use calculus to find the exact value of k for which $\mathrm{J}(k)$ is least, and show that the least value of $\mathrm{J}(k)$ is $\ln\frac{27}{16}$. [4]

(d) The function g is continuous and increasing for $1 \leq x \leq 3$. Without doing any detailed calculation, state, in terms of g, the value of k for which $\int_1^3 |\mathrm{g}(x) - k| \, \mathrm{d}x$ is least, and explain your answer. [3]

Solution:

(a) Integrate by parts with $u = \ln x$, $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$:

$\int \ln x \, \mathrm{d}x = x\ln x - \int x \cdot \frac{1}{x} \, \mathrm{d}x = x\ln x - x + c$. [M1 by parts with $\frac{\mathrm{d}v}{\mathrm{d}x} = 1$; A1]

(b) The line $y = k$ meets the curve $y = \ln x$ where $x = \mathrm{e}^k$. Since $0 \leq k \leq \ln 3$, $1 \leq \mathrm{e}^k \leq 3$, so the split point lies in $[1,3]$ and MOVES with k.

For $1 \leq x \leq \mathrm{e}^k$, $\ln x \leq k$, so $|\ln x - k| = k - \ln x$; for $\mathrm{e}^k \leq x \leq 3$, $|\ln x - k| = \ln x - k$. [M1: the split at $x = \mathrm{e}^k$ with the correct sign on each side]

$\mathrm{J}(k) = \int_1^{\mathrm{e}^k} (k - \ln x) \, \mathrm{d}x + \int_{\mathrm{e}^k}^3 (\ln x - k) \, \mathrm{d}x$. [M1: both integrals evaluated with the result of (a) and the limits e^k]

First integral: $[kx - x\ln x + x]_1^{\mathrm{e}^k} = (k\mathrm{e}^k - k\mathrm{e}^k + \mathrm{e}^k) - (k - 0 + 1) = \mathrm{e}^k - k - 1$. [A1]

Second integral: $[x\ln x - x - kx]_{\mathrm{e}^k}^3 = (3\ln 3 - 3 - 3k) - (k\mathrm{e}^k - \mathrm{e}^k - k\mathrm{e}^k) = 3\ln 3 - 3 - 3k + \mathrm{e}^k$.

$\mathrm{J}(k) = 2\mathrm{e}^k - 4k + 3\ln 3 - 4$. [A1]

(Checks: $\mathrm{J}(0) = 3\ln 3 - 2 = \int_1^3 \ln x \, \mathrm{d}x$; $\mathrm{J}(\ln 3) = 2 - \ln 3 = \int_1^3 (\ln 3 - \ln x) \, \mathrm{d}x$. A candidate who ignores the modulus gets $3\ln 3 - 2 - 2k$, which has no stationary value, so part (c) exposes the error.)

(c) $\mathrm{J}'(k) = 2\mathrm{e}^k - 4$. [M1: differentiate the answer to (b) with respect to k and set it equal to 0]

$\mathrm{J}'(k) = 0 \Rightarrow \mathrm{e}^k = 2 \Rightarrow k = \ln 2$, which lies in $[0, \ln 3]$. [A1]

$\mathrm{J}''(k) = 2\mathrm{e}^k > 0$, so $k = \ln 2$ gives a minimum; as it is the only stationary value in $[0, \ln 3]$ ($\mathrm{J}'(k) < 0$ for $k < \ln 2$ and $\mathrm{J}'(k) > 0$ for $k > \ln 2$), it is the least value of J. [B1]

$\mathrm{J}(\ln 2) = 2(2) - 4\ln 2 + 3\ln 3 - 4 = 3\ln 3 - 4\ln 2 = \ln 27 - \ln 16 = \ln\frac{27}{16}$ (shown). [A1]

(Numerically $\ln\frac{27}{16} \approx 0.523$, against $\mathrm{J}(0) \approx 1.30$ and $\mathrm{J}(\ln 3) \approx 0.901$.)

(d) $k = \mathrm{g}(2)$. [B1]

$\int_1^3 |\mathrm{g}(x) - k|\ \mathrm{d}x$ is the total area between the curve $y = \mathrm{g}(x)$ and the line $y = k$ for $1 \le x \le 3$. As g is increasing, a line with $\mathrm{g}(1) \le k \le \mathrm{g}(3)$ meets the curve at exactly one point, $x = c$, and is above the curve on $[1, c]$ and below it on $[c, 3]$.

Raising the line by a small amount δ adds a strip of height δ over $[1, c]$ to the left-hand area (about $\delta(c - 1)$) and removes a strip of height δ over $[c, 3]$ from the right-hand area (about $\delta(3 - c)$), a net change of about $\delta[(c - 1) - (3 - c)] = 2\delta(c - 2)$. [B1]

So the total area decreases as the line rises while $c < 2$ and increases once $c > 2$: it is least when the line meets the curve at the midpoint $x = 2$ of the interval, i.e. when $k = \mathrm{g}(2)$. [B1]

(A line wholly below the curve, $k < \mathrm{g}(1)$, reduces the area by about 2δ when raised, and a line wholly above it, $k > \mathrm{g}(3)$, reduces it when lowered, so the least value has the crossing inside [1,3].)

Consistent with (c): for $\mathrm{g}(x) = \ln x$ the crossing is $c = \mathrm{e}^k$ and $\mathrm{J}'(k) = 2\mathrm{e}^k - 4 = 2(c - 2)$, exactly the rate found above, and $k = \ln 2 = \mathrm{g}(2)$. The numbers of (c) point to $\mathrm{g}(2)$ and to nothing else a candidate might guess: $\frac{1}{2}[\mathrm{g}(1) + \mathrm{g}(3)] = \frac{1}{2}\ln 3 \approx 0.549$ and the mean value $\frac{1}{2}\int_1^3 \ln x\ \mathrm{d}x = \frac{3\ln 3 - 2}{2} \approx 0.648$, whereas $\ln 2 \approx 0.693$.

Mark scheme notes. (b) Any equivalent form earns the final A1, e.g. $2\mathrm{e}^k - 4k + \ln 27 - 4$. (c) The B1 needs a nature test (second derivative or sign of J′); a stationary value found without one scores M1 A1 B0 A1. (d) The value alone, with no reason, scores B1 only. Accept, for the explanation, the equivalent statement that at the least value the length of [1,3] on which $\mathrm{g}(x) < k$ must equal the length on which $\mathrm{g}(x) > k$, because a small rise in k changes the area at the rate (first length) − (second length); accept 'the line $y = k$ meets the curve at $x = 2$' for the value. A candidate who differentiates a general $\int_1^c (k - \mathrm{g})\ \mathrm{d}x + \int_c^3 (\mathrm{g} - k)\ \mathrm{d}x$ with respect to k (the terms from the moving limit vanish because the integrand is zero at $x = c$) earns full marks, but this is not expected.

10. A ball of ice is melting in a warm room and remains spherical as it melts. At time t hours after it is first observed, its radius is r cm and its volume is V cm^3; the radius is 10 cm when $t = 0$ and 8 cm when $t = 1$. Two models are proposed for the melting: in model A, the rate of decrease of V is proportional to the surface area of the ball, and in model B, the rate of decrease of V is proportional to V.

[The volume of a sphere of radius r is $\frac{4}{3}\pi r^3$ and its surface area is $4\pi r^2$.]

(a) Using model A, find r in terms of t. Hence state the time at which, according to model A, the ball has melted completely. [4]

(b) Using model B, find r in terms of t. [3]

(c) When $t = 3$, the radius of the ball is measured as 4.2 cm. Determine which model fits this measurement better. [2]

(d) Explain why model B cannot describe the whole of the melting. [1]

(e) Show that, for $0 < t < 1$, model B predicts a smaller radius than model A. [3]

Solution:

(a) Model A: the rate of decrease of V is proportional to the surface area $4\pi r^2$, so $\frac{\mathrm{d}V}{\mathrm{d}t} = -k(4\pi r^2)$, where k is a positive constant. M1

Since $V = \frac{4}{3}\pi r^3$, by the chain rule $\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}r} \times \frac{\mathrm{d}r}{\mathrm{d}t} = 4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t}$.

So $4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t} = -4\pi k r^2$, and since $r > 0$ while the ball exists, $\frac{\mathrm{d}r}{\mathrm{d}t} = -k$: the surface area cancels and the radius decreases at a constant rate. M1

Integrating, $r = -kt + C$. When $t = 0$, $r = 10$, so $C = 10$; when $t = 1$, $r = 8$, so $8 = 10 - k$ and $k = 2$. $r = 10 - 2t$. A1

The ball has melted completely when $r = 0$: $10 - 2t = 0$, so $t = 5$, i.e. 5 hours after it is first observed (model A applies for $0 \le t \le 5$). B1

(Alternative for the first two marks: $S = 4\pi r^2 = (36\pi)^{\frac{1}{3}} V^{\frac{2}{3}}$, so $\frac{\mathrm{d}V}{\mathrm{d}t} = -k(36\pi)^{\frac{1}{3}} V^{\frac{2}{3}}$; separating, $3V^{\frac{1}{3}} = -k(36\pi)^{\frac{1}{3}} t + D$, so $V^{\frac{1}{3}}$, and hence r, is a linear function of t. Same marks.)

(b) Model B: $\frac{\mathrm{d}V}{\mathrm{d}t} = -cV$, where c is a positive constant. M1

Separating the variables: $\int \frac{1}{V}\,\mathrm{d}V = -\int c\,\mathrm{d}t$, so $\ln V = -ct + D$ ($V > 0$) and $V = V_0 \mathrm{e}^{-ct}$, where $V_0 = \frac{4}{3}\pi(10)^3$ is the volume when $t = 0$. M1

Then $\frac{4}{3}\pi r^3 = \frac{4}{3}\pi(1000)\mathrm{e}^{-ct}$, so $r^3 = 1000\mathrm{e}^{-ct}$ and $r = 10\mathrm{e}^{-\frac{ct}{3}}$. M1

(Alternatively, in r directly: $4\pi r^2 \frac{\mathrm{d}r}{\mathrm{d}t} = -c \cdot \frac{4}{3}\pi r^3$, so $\frac{\mathrm{d}r}{\mathrm{d}t} = -\frac{c}{3}r$ and $r = 10\mathrm{e}^{-\frac{ct}{3}}$.)

When $t = 1$, $r = 8$: $\mathrm{e}^{-\frac{c}{3}} = 0.8$ (so $c = 3\ln 1.25 = 0.669$).

$r = 10\left(\mathrm{e}^{-\frac{c}{3}}\right)^t = 10(0.8)^t$. A1

(Accept $r = 10\mathrm{e}^{t\ln 0.8}$ or $r = 10\mathrm{e}^{-0.223t}$.)

(c) When $t = 3$: model A gives $r = 10 - 2(3) = 4$; model B gives $r = 10(0.8)^3 = 5.12$. M1

$|4.2 - 4| = 0.2$ and $|4.2 - 5.12| = 0.92$. Since $0.2 < 0.92$, model A fits the measurement better. A1

(Comparing volumes leads to the same conclusion.)

(d) In model B, $r = 10(0.8)^t > 0$ for every value of t, and $r \to 0$ only as $t \to \infty$. So model B predicts that the ball never melts completely, whereas a ball of ice in a warm room melts completely in a finite time (model A has it gone at $t = 5$). So model B cannot describe the final stage of the melting. B1

(e) Let $h(t) = (10 - 2t) - 10(0.8)^t$, the radius predicted by model A minus the radius predicted by model B, for $0 \le t \le 1$.

Both models were fitted to the same two readings, so $h(0) = 10 - 10 = 0$ and $h(1) = 8 - 8 = 0$.

$h'(t) = -2 - 10(\ln 0.8)(0.8)^t$ and $h''(t) = -10(\ln 0.8)^2(0.8)^t < 0$ for all t. M1

So the graph of $y = h(t)$ is concave downward on $0 \le t \le 1$, and therefore lies above its chord joining $(0,0)$ and $(1,0)$, which is the t-axis. Hence $h(t) > 0$ for $0 < t < 1$, i.e. $10(0.8)^t < 10 - 2t$: model B predicts a smaller radius than model A. A1

(Equivalent argument: h' is strictly decreasing, with $h'(0) = -2 - 10\ln 0.8 = 0.231 > 0$ and $h'(1) = -2 - 8\ln 0.8 = -0.215 < 0$, so h increases to a single maximum and then decreases on $[0,1]$; with $h(0) = h(1) = 0$ this gives $h(t) > 0$ for $0 < t < 1$. Or: $\frac{\mathrm{d}^2}{\mathrm{d}t^2}[10(0.8)^t] = 10(\ln 0.8)^2(0.8)^t > 0$, so model B's curve is concave upward, and model A's line $r = 10 - 2t$ is its chord through $(0,10)$ and $(1,8)$, so the curve lies below the line between these points.)

Marker's notes: checking one or several values (e.g. $t = 0.5$: 9 against 8.944) or reading a graphing-calculator plot scores 0, since the difference is at most 0.0557 cm (at $t = 0.491$) and a value check proves nothing about every t. The final mark needs both $h(0) = h(1) = 0$ and the sign of h'' (or the sign change of h').

(Remark: the same concavity gives $h(t) < 0$ for $1 < t \le 5$ — model B predicts the LARGER radius after the second reading, as found at $t = 3$ in part (c). The two models never differ by more than 0.056 cm during the first hour, which is why they can be told apart only by a reading taken later.)

Mark scheme note (moderation, 27 Sep 2026): re-marked (b) [3] and (e) [3] — (e) is the question's proof (h(0) = h(1) = 0 with h" < 0, or the convexity of 0.8^t against its chord), (b) the standard separation with one constant.

11. A solar panel $ABCD$, in the shape of a rectangle, is fixed to the flat roof of a building. Coordinates (x, y, z), in metres, are measured from an origin O on the roof, with the x-axis pointing east, the y-axis pointing north and the z-axis pointing vertically upwards, so that the roof lies in the plane $z = 0$. The lower edge AB of the panel rests on the roof, and the points A, B and D have coordinates $(4,1,0)$, $(8,3,0)$ and $(3,3,1)$ respectively. A vertical mast stands on the roof at the point $(4,0,0)$, and its top T is 5 m above the roof. The thicknesses of the panel and of the mast are negligible.

(a) Find a cartesian equation of the plane containing the panel. [4]

(b) Find the acute angle between the panel and the roof. [2]

(c) The sun's rays are modelled as parallel straight lines, and the power produced by the panel is proportional to $\sin\alpha$, where α is the acute angle between the sun's rays and the panel. At 9 a.m. on a certain day, the sun's rays travel in the direction $\begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}$. Find the power produced by the panel at this time as a percentage of the greatest power that the panel can produce. [2]

(d) At a certain time on another day, the panel produces the greatest power that it can. Find the coordinates of the point S where the sun's ray through T meets the plane containing the panel at this time. [4]

(e) Determine whether the shadow of T falls on the panel at this time. [2]

Solution:

(a) $\overrightarrow{AB} = \begin{pmatrix}4\\2\\0\end{pmatrix}, \overrightarrow{AD} = \begin{pmatrix}-1\\2\\1\end{pmatrix}$.

A normal to the panel is $\overrightarrow{AB} \times \overrightarrow{AD} = \begin{pmatrix}2(1)-0(2)\\0(-1)-4(1)\\4(2)-2(-1)\end{pmatrix} = \begin{pmatrix}2\\-4\\10\end{pmatrix} = 2\begin{pmatrix}1\\-2\\5\end{pmatrix}$, so take $\boldsymbol{n} = \begin{pmatrix}1\\-2\\5\end{pmatrix}$.

$\boldsymbol{r}\cdot\boldsymbol{n} = \overrightarrow{OA}\cdot\boldsymbol{n} = 4-2+0 = 2$, so the plane is $x-2y+5z=2$.

✓ **Check:** B gives $8-6+0=2$ and D gives $3-6+5=2$.) [M1 two direction vectors in the plane, M1 vector product, M1 $\boldsymbol{r}\cdot\boldsymbol{n} = \boldsymbol{a}\cdot\boldsymbol{n}$, A1]

(b) The roof $z=0$ has normal $\boldsymbol{k} = \begin{pmatrix}0\\0\\1\end{pmatrix}$.

$\cos\theta = \frac{|\boldsymbol{n}\cdot\boldsymbol{k}|}{|\boldsymbol{n}||\boldsymbol{k}|} = \frac{5}{\sqrt{30}}$, so $\theta = 24.1°$ (24.095...°). [M1, A1]

(c) Let $\boldsymbol{d} = \begin{pmatrix}-1\\0\\-1\end{pmatrix}$. The angle between a line and a plane: $\sin\alpha = \frac{|\boldsymbol{d}\cdot\boldsymbol{n}|}{|\boldsymbol{d}||\boldsymbol{n}|} = \frac{|-1+0-5|}{\sqrt{2}\sqrt{30}} = \frac{6}{\sqrt{60}} = \frac{\sqrt{15}}{5} = 0.7746$ ($\alpha = 50.8°$).

The power is greatest when $\sin\alpha = 1$ (the rays perpendicular to the panel), so the power at 9 a.m. as a percentage of the greatest power is $100\sin\alpha = 77.5\%$. [M1 $\sin\alpha$ from $|\boldsymbol{d}\cdot\boldsymbol{n}|$, A1]

(d) The greatest power needs $\sin\alpha = 1$, i.e. $\alpha = 90°$: the sun's rays are perpendicular to the panel, so they are parallel to $\boldsymbol{n}$ (travelling in the direction $-\boldsymbol{n}$, downwards).

T is $(4,0,5)$. The ray through T is $\boldsymbol{r} = \begin{pmatrix}4\\0\\5\end{pmatrix} + s\begin{pmatrix}1\\-2\\5\end{pmatrix}$.

Substituting into $x-2y+5z=2$: $(4+s)-2(-2s)+5(5+5s)=2 \Rightarrow 29+30s=2 \Rightarrow s=-\frac{9}{10}$.

So $S = (4-0.9,\ 0+1.8,\ 5-4.5) = (3.1,\ 1.8,\ 0.5)$. ($S$ is the foot of the perpendicular from T to the plane; the ray goes down from height 5 m to height 0.5 m, so it meets the plane before the roof.) [B1 rays parallel to $\boldsymbol{n}$ (deduced from $\sin\alpha = 1$), M1 line through T, M1 substitute into the plane, A1]

(e) The points of the panel are $\overrightarrow{OA} + \lambda\overrightarrow{AB} + \mu\overrightarrow{AD}$ with $0 \le \lambda \le 1$ and $0 \le \mu \le 1$.

$\overrightarrow{AS} = \begin{pmatrix}-0.9\\0.8\\0.5\end{pmatrix} = \lambda\begin{pmatrix}4\\2\\0\end{pmatrix} + \mu\begin{pmatrix}-1\\2\\1\end{pmatrix}$: the z-components give $\mu = 0.5$ and the x-components give $4\lambda - 0.5 = -0.9$, so $\lambda = -0.1$ (check y: $2(-0.1)+2(0.5)=0.8$).

(Equivalently, since $AB \perp AD$: $\lambda = \frac{\overrightarrow{AS}\cdot\overrightarrow{AB}}{|\overrightarrow{AB}|^2} = \frac{-2}{20} = -0.1$ and $\mu = \frac{\overrightarrow{AS}\cdot\overrightarrow{AD}}{|\overrightarrow{AD}|^2} = \frac{3}{6} = 0.5$.)

Since $\lambda = -0.1 < 0$, S lies outside the rectangle, beyond the edge AD (by $0.1|\overrightarrow{AB}| = 0.447$ m). So the shadow of T does not fall on the panel at this time. [M1 λ and μ found, A1 conclusion with the reason $\lambda < 0$]

Marker's note: S lies within the ranges of the corners' coordinates (C is $(7,5,1)$, so $3 \le x \le 8$, $1 \le y \le 5$, $0 \le z \le 1$), so comparing coordinates with the corners does not decide the question; an answer 'yes' from that comparison, or from S lying on the plane, scores 0.

Answers

1 (a) $a = \frac{-1 \pm \sqrt{21}}{2}$; (b) $\frac{\sqrt{21}-25}{2}$

2 (a) Shown. (b) $n = 1,3,5,7$

3 (a) $\mathrm{f}'(x) = 0$ (shown); (b) $\mathrm{R_f} = \left\{-\frac{2\pi}{3}, \frac{\pi}{3}\right\}$

4 (a) B: $4 + \mathrm{i}$, D: $2 + 5\mathrm{i}$; (b) $p = 3 + 3\mathrm{i}$, $q = -7 + 24\mathrm{i}$ (shown); (c) $w = -2 - \mathrm{i},\ 1 - 2\mathrm{i},\ 2 + \mathrm{i},\ -1 + 2\mathrm{i}$; $s = \pm(3 + 4\mathrm{i})$

5 (a) $(-\infty, -2] \cup [2, \infty)$; (b) $x = \frac{3-\sqrt{5}}{2}$, $x = \frac{3+\sqrt{5}}{2}$; (c) $k < -10$

6 (a) $x = 1$ minimum, $x = 4$ maximum (sign change of f'); (b) greatest 10, least 1; (c) $(2, -1)$ maximum, $(8, -17)$ minimum

7 (a) shown; (b) $\theta \approx \sqrt{6k}$; (c) $\frac{a(1-\cos\theta)}{\sin\theta}\left(= a\tan\frac{1}{2}\theta\right) \approx \frac{1}{2}a\theta$; (d) 0.474 m (about 47 cm): nearly 50 times the 1 cm expansion, so a tiny expansion makes a large bow and rails need room to expand.

8 (a) shown; (b) $h = \frac{\sqrt{2}}{2}$, $B\left(\frac{3\pi}{4} + \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$; (c) $\frac{3\sqrt{2}-4}{4}$

9 (a) $x\ln x - x + c$; (b) $\mathrm{J}(k) = 2\mathrm{e}^k - 4k + 3\ln 3 - 4$; (c) $k = \ln 2$, least value $\ln\frac{27}{16}$ (shown); (d) $k = \mathrm{g}(2)$, with the explanation that raising the line changes the total area at the rate (length where the line is above the curve) − (length where it is below), which is zero when the line crosses the curve at the midpoint $x = 2$

10 (a) $r = 10 - 2t$; melted completely at $t = 5$ (5 hours); (b) $r = 10(0.8)^t$; (c) A predicts 4 cm, B predicts 5.12 cm, so model A fits better ($0.2 < 0.92$); (d) $10(0.8)^t > 0$ for all t, so in model B the ball never melts completely; (e) $h(t) = 10 - 2t - 10(0.8)^t$ has $h(0) = h(1) = 0$ and $h'' < 0$, so $h(t) > 0$ for $0 < t < 1$.

11 (a) $x - 2y + 5z = 2$; (b) 24.1°; (c) 77.5%; (d) $S(3.1,\ 1.8,\ 0.5)$; (e) No: $\overrightarrow{AS} = -0.1\,\overrightarrow{AB} + 0.5\,\overrightarrow{AD}$ with $-0.1 < 0$, so S is outside the panel (beyond edge AD).